AI-Powered Smart Campus Analytics & Early Risk Detection System

Executive Analytics Report

CS Department | 500 Students | March 2025

1. Key Performance Indicators

500 Total Students	**71** At-Risk Students	**14** Critical Risk
140.5 Avg Semester Marks	**71.9%** Avg Attendance	**31** Grade A Students

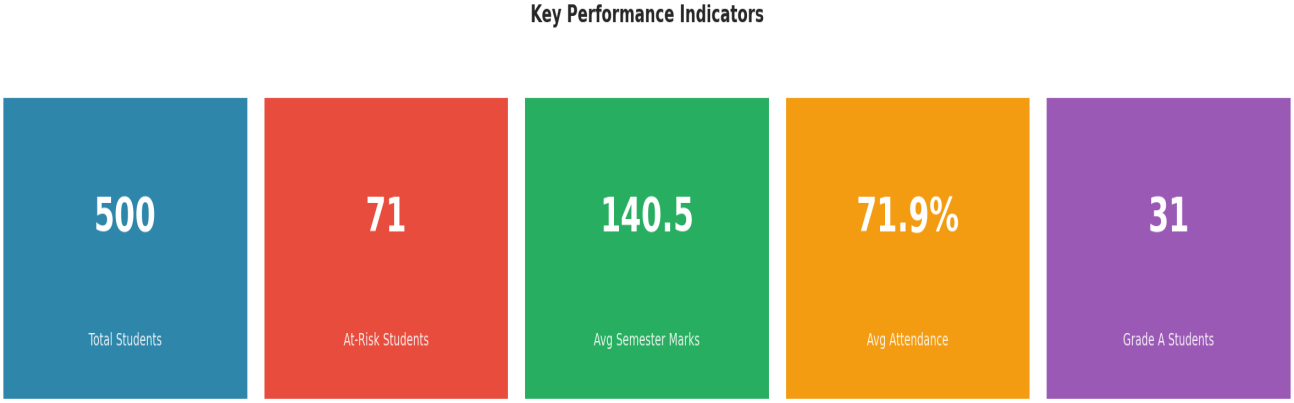

Figure 1: KPI Summary Dashboard

2. Grade & Risk Distribution

2.1 Grade Distribution

Grade	Count	Percentage	Marks Range
A	31	6.2%	165–200

B	112	22.4%	150–165
C	170	34.0%	135–150
D	128	25.6%	120–135
F	59	11.8%	<120

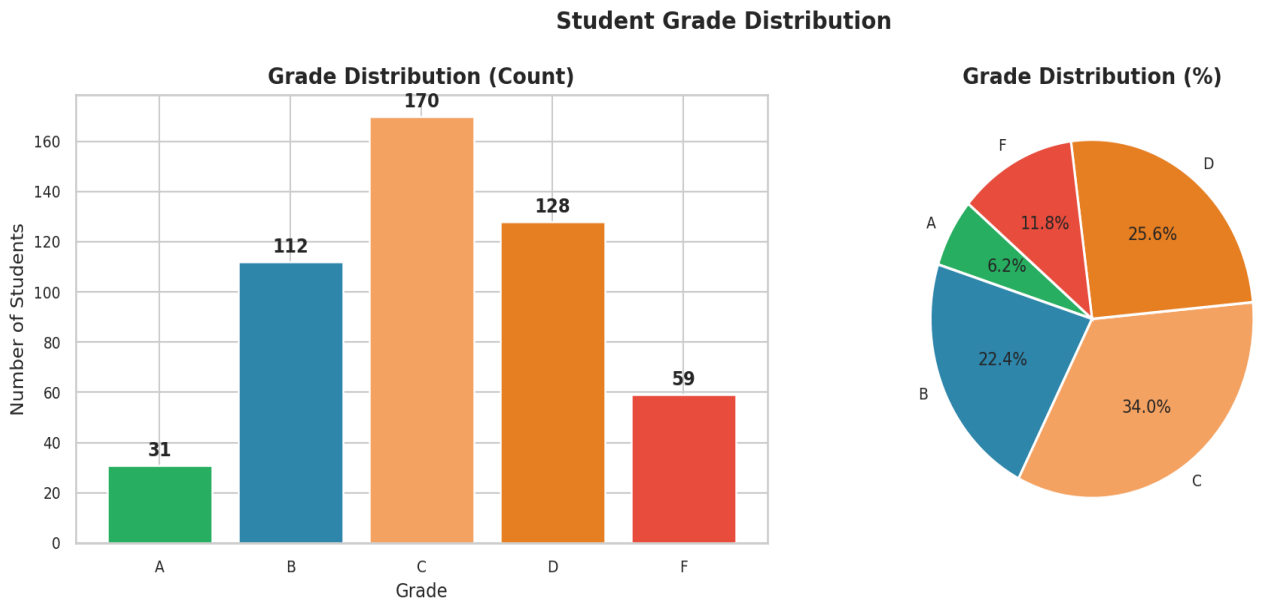

Figure 2: Student Grade Distribution

2.2 Risk Tier Distribution

Risk Tier	Count	Percentage	Required Action
Low	236	47.2%	No action — monitor regularly
Moderate	193	38.6%	Faculty advisory recommended
High	57	11.4%	Mandatory counselling & attendance review
Critical	14	2.8%	IMMEDIATE intervention required

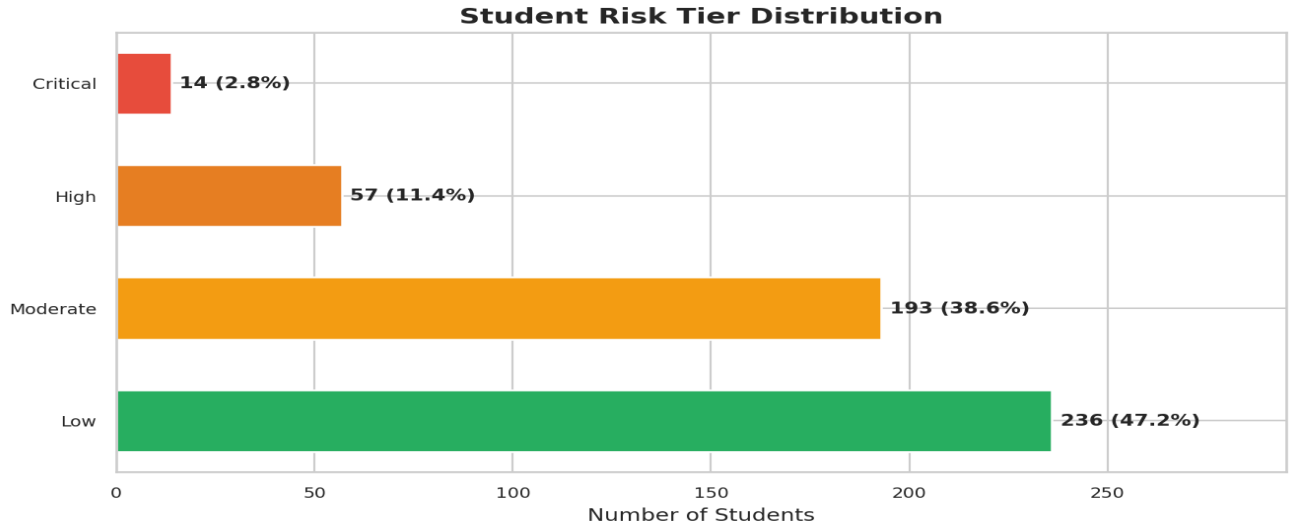

Figure 3: Risk Tier Distribution

3. Exploratory Data Analysis

Figure 4: Feature Distributions Across All Students

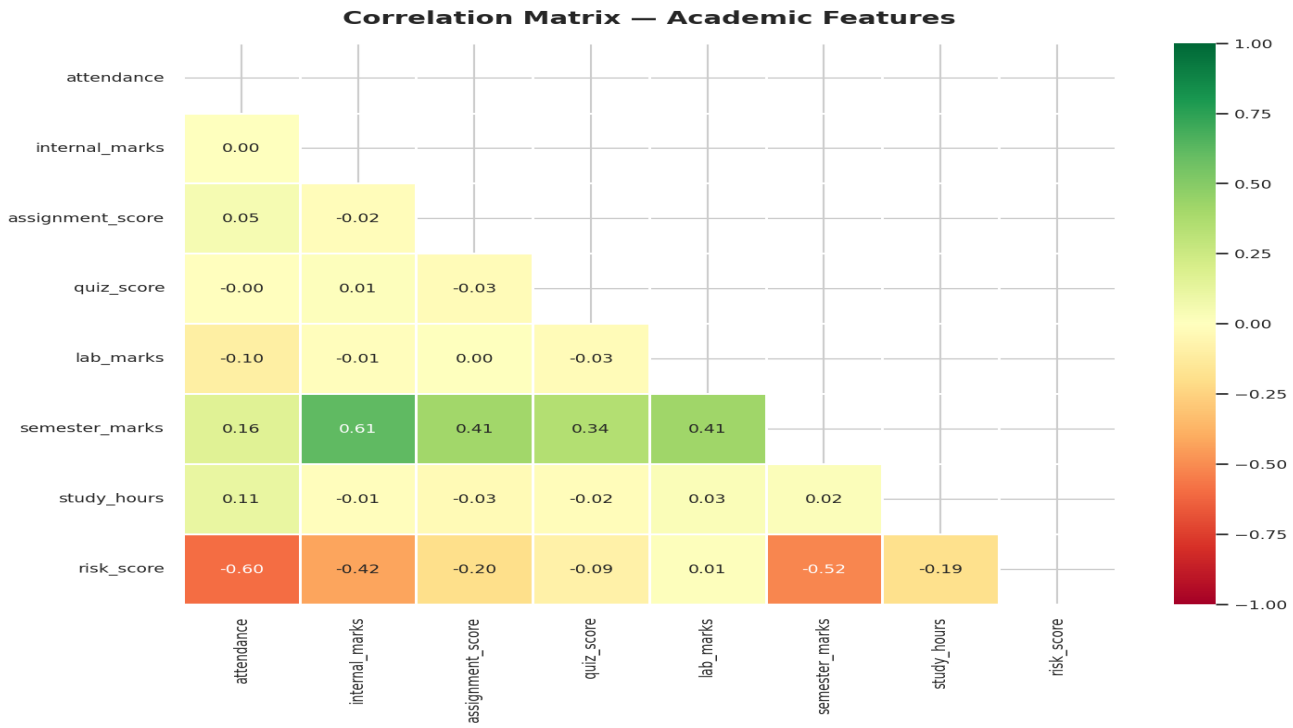

Figure 5: Correlation Matrix — Academic Features

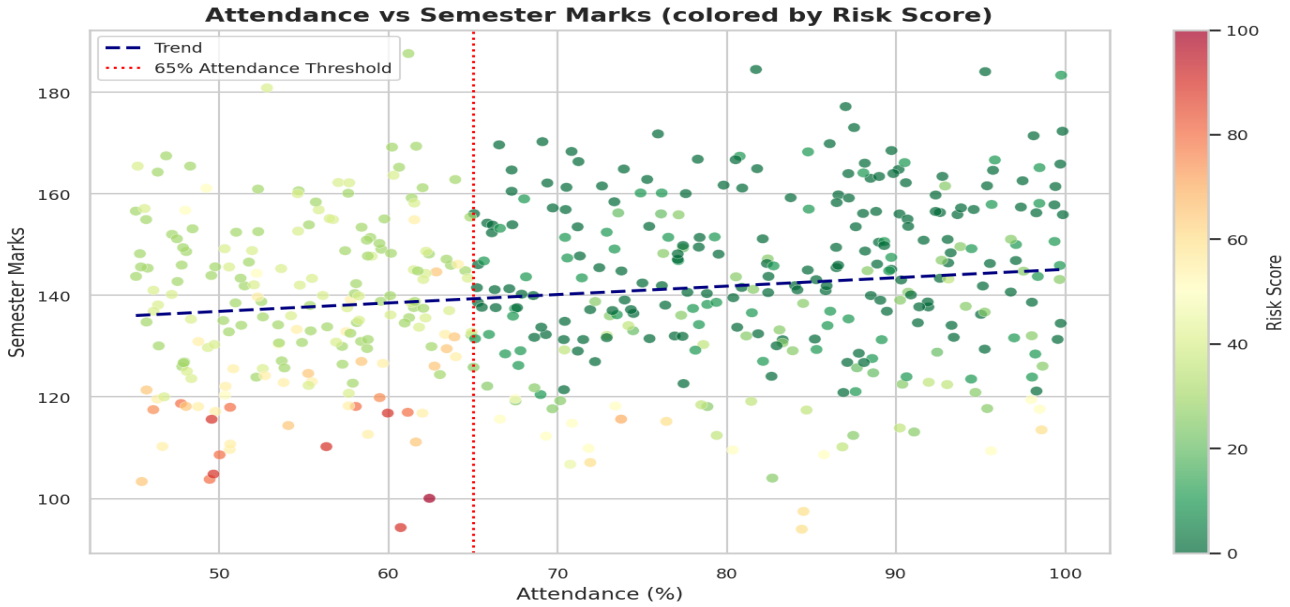

Figure 6: Attendance vs Semester Marks (colored by Risk Score)

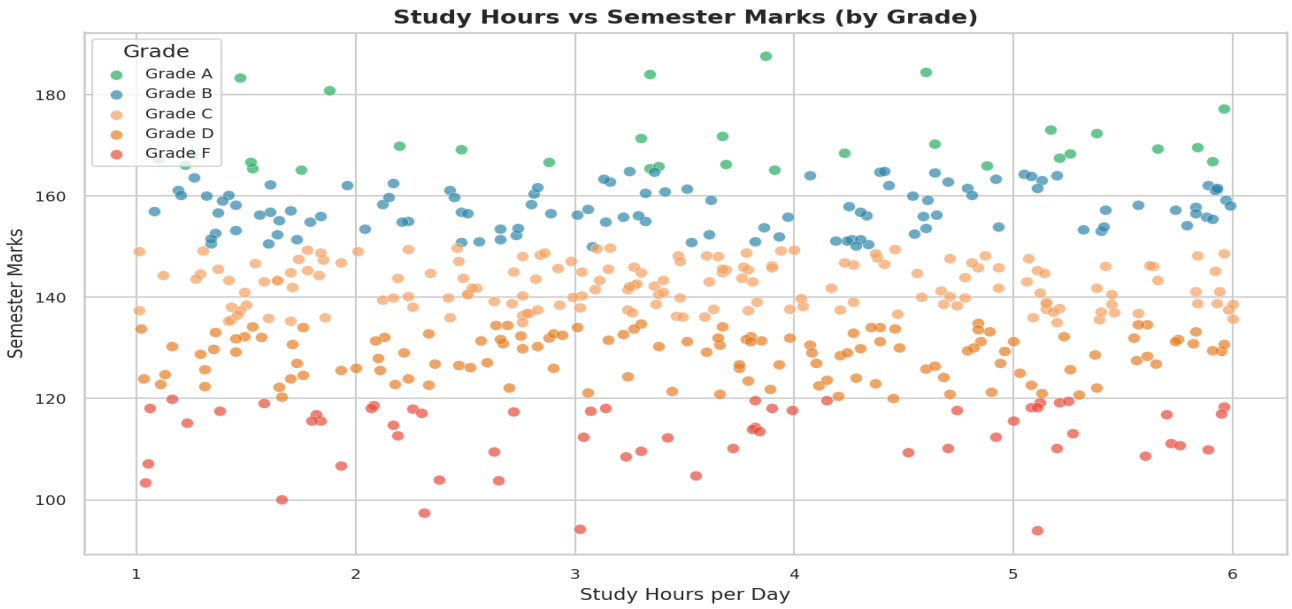

Figure 7: Study Hours vs Semester Marks by Grade

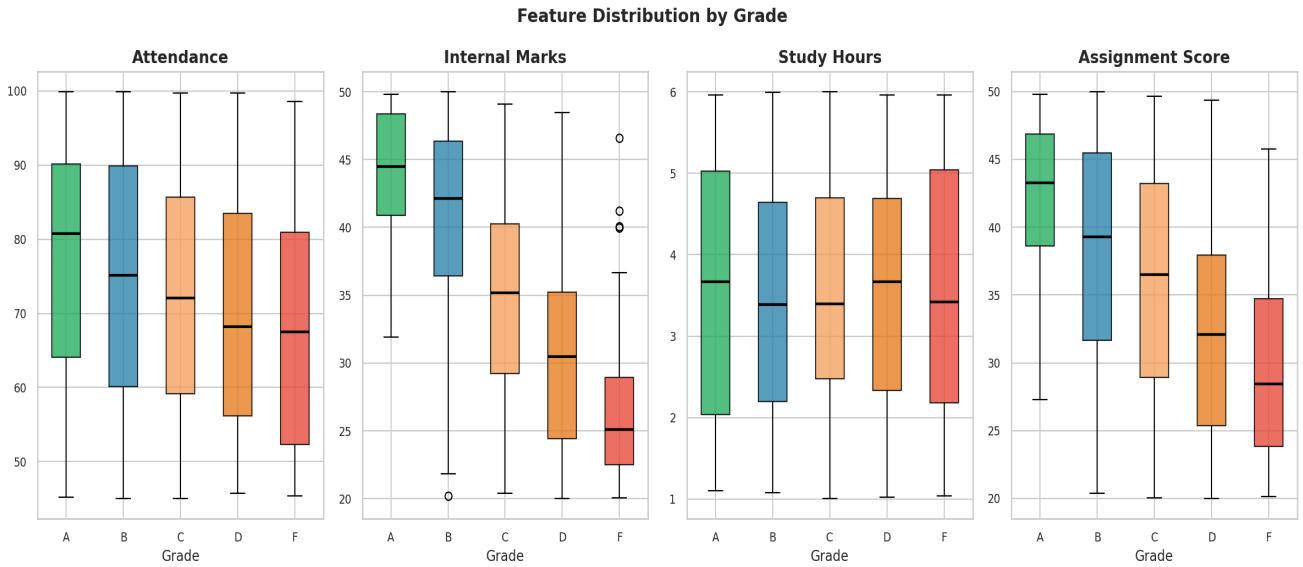

Figure 8: Feature Distribution by Grade Group

4. Machine Learning Model Results

4.1 Model Performance Summary

Model Type	Model Name	Key Metric	Score	Status
Regression	**Linear Regression**	**R^2 Score**	**0.8743**	**Best**
Regression	Random Forest	R^2 Score	0.7631	
Regression	Gradient Boosting	R^2 Score	0.8210	
Classification	**Logistic Regression**	**Accuracy**	**63.0%**	**Best**
Classification	Decision Tree	Accuracy	40.0%	
Classification	Random Forest	Accuracy	54.0%	

Figure 9: Model Performance Comparison

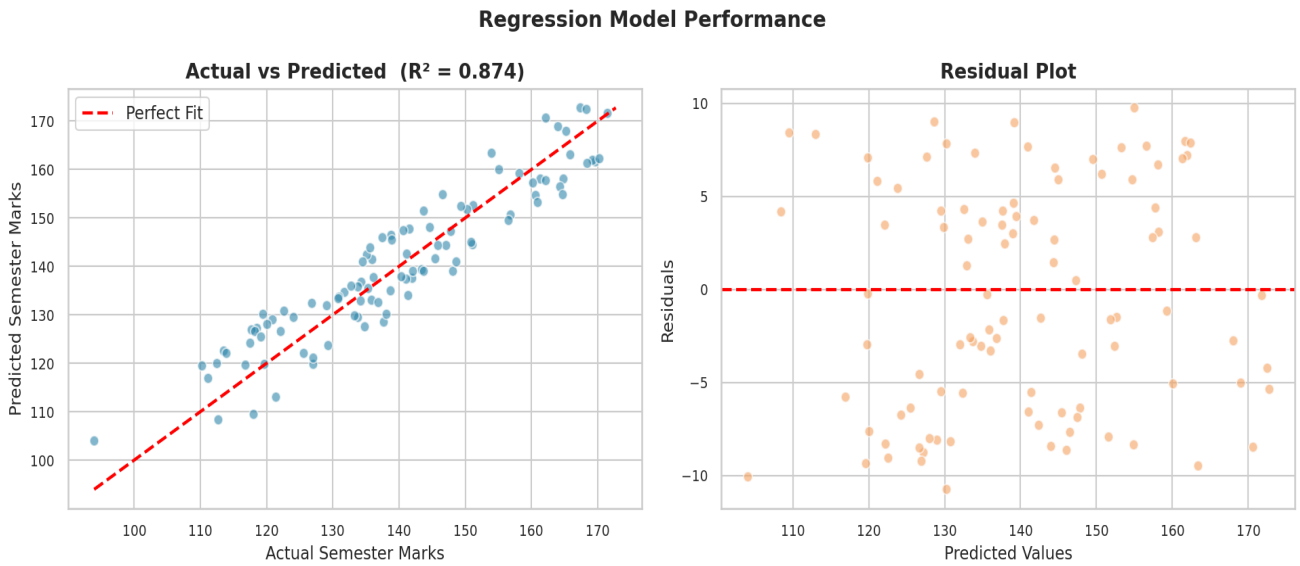

Figure 10: Regression Results — Actual vs Predicted

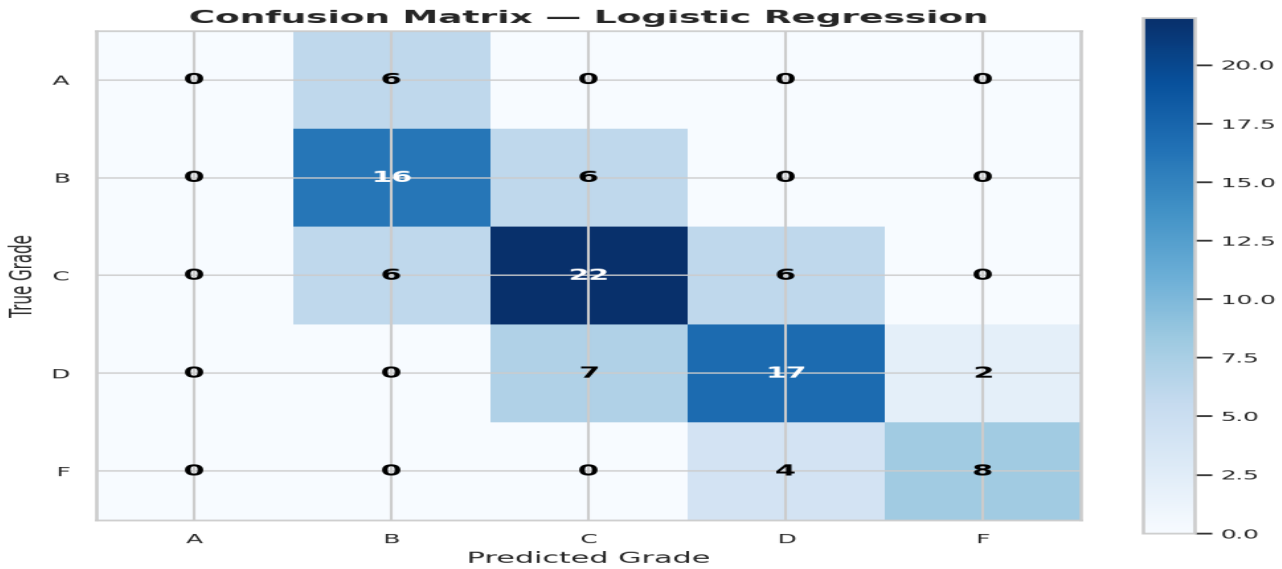

Figure 11: Classification Confusion Matrix (Best Model)

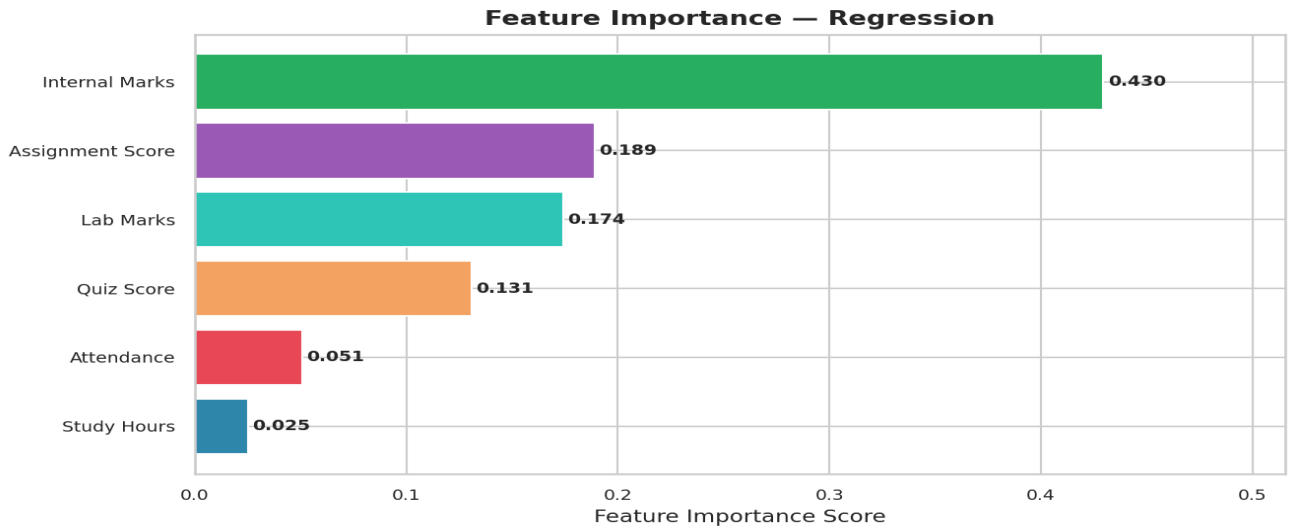

Figure 12: Feature Importance — Regression

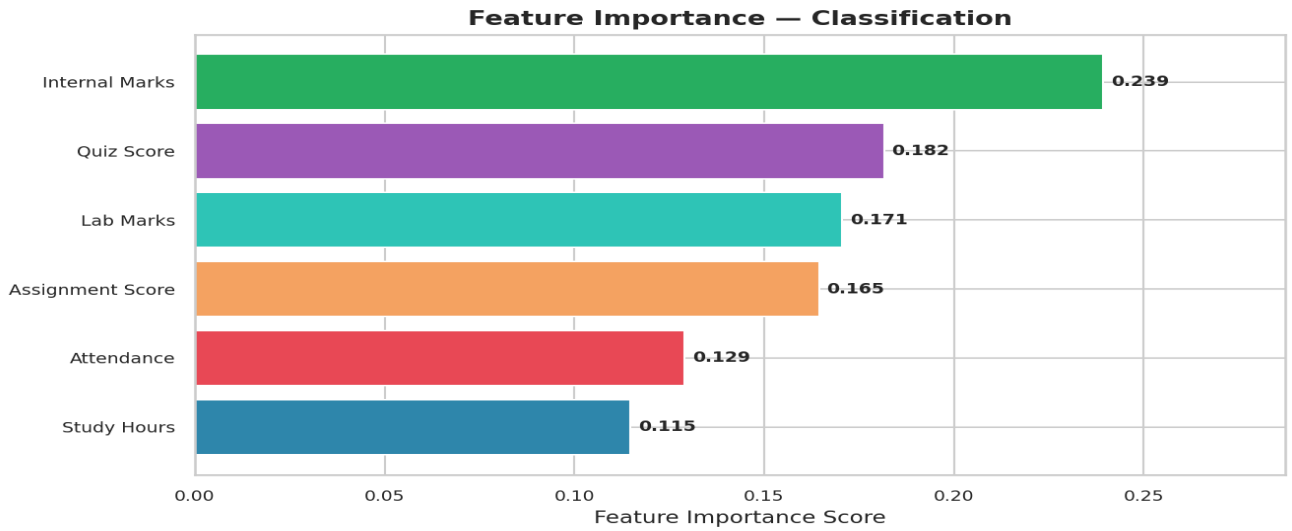

Figure 13: Feature Importance — Classification

5. Student Clustering & Personas

K-Means clustering identified **4 distinct student groups** based on attendance, marks, study hours, and assignment performance.

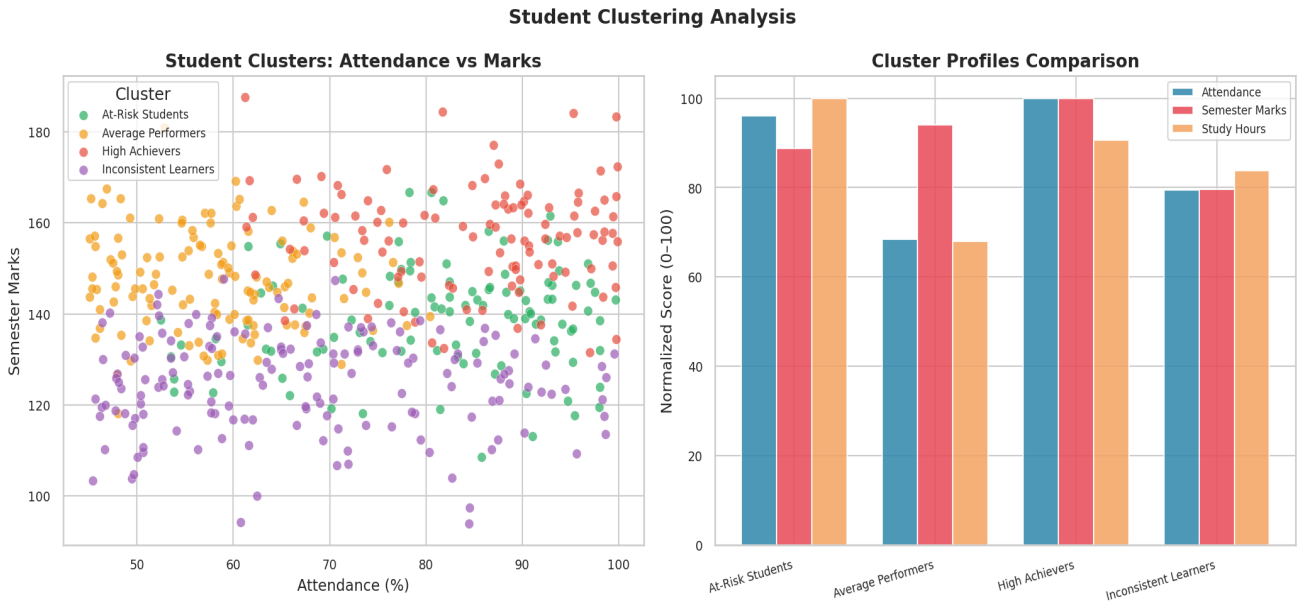

Figure 14: Student Clustering Analysis

Cluster Name	Avg Attendance	Avg Semester Marks	Avg Study Hrs	Count
At-Risk Students	81.5%	139.2	4.14h	106
Average Performers	58.0%	147.3	2.81h	124
High Achievers	84.8%	156.7	3.75h	111
Inconsistent Learners	67.4%	124.6	3.47h	159

6. At-Risk Students — Early Warning List

71 students are currently flagged as at-risk (**14 Critical**). The table below shows the top 20 by risk score. Full details are available in the Excel report.

USN	Name	Attend%	Internal	Sem.Marks	Risk Score	Tier	Grade
1RV21CS268	Karthik Bose	62.4%	23.2	100.0	100	Critical	F
1RV21CS336	Yash Pillai	60.0%	22.2	116.8	90	Critical	F
1RV21CS239	Vivaan Kapoor	56.3%	23.0	110.2	90	Critical	F
1RV21CS290	Vikram Patil	49.7%	24.0	104.8	90	Critical	F
1RV21CS294	Mahesh Hegde	49.5%	20.9	115.6	90	Critical	F
1RV21CS202	Dev Varma	60.7%	20.2	94.3	90	Critical	F
1RV21CS176	Gokul Reddy	61.1%	21.3	117.0	80	Critical	F
1RV21CS309	Arvind Tiwari	50.0%	22.9	108.6	80	Critical	F
1RV21CS079	Vivek Chatterjee	50.6%	20.4	118.0	80	Critical	F
1RV21CS140	Tanmay Saxena	47.7%	22.4	118.7	80	Critical	F
1RV21CS365	Sanjay Mishra	58.1%	21.5	118.1	80	Critical	F
1RV21CS057	Yash Chavan	49.4%	24.5	103.8	80	Critical	F
1RV21CS362	Aditya Dubey	46.1%	27.8	117.5	75	Critical	F
1RV21CS465	Prakash Naidu	59.5%	46.6	119.9	75	Critical	F
1RV21CS436	Arvind Sharma	73.8%	20.8	115.6	70	High	F
1RV21CS129	Arjun Nair	62.7%	24.4	126.1	65	High	D
1RV21CS142	Tarun Patel	45.4%	29.0	103.3	65	High	F
1RV21CS061	Akash Paul	54.1%	34.0	114.4	65	High	F
1RV21CS476	Ajay Verma	58.4%	23.8	127.0	65	High	D
1RV21CS310	Aarav Shekhar	63.9%	23.6	131.8	65	High	D

7. Recommendations & Action Plan

Immediate Interventions

• Contact all 14 Critical-risk students and their guardians immediately.

• Assign dedicated academic mentors to High and Critical risk students.

• Review attendance records for students below 65% — initiate attendance improvement plan.

Faculty Actions

• Conduct weekly attendance spot-checks for at-risk groups.

• Introduce supplementary tutorial sessions for students with internal marks below 25.

• Share personalized performance reports with students after mid-semester.

System Improvements

• Integrate this analytics system with the college ERP for real-time monitoring.

• Retrain ML models each semester with updated data for improved accuracy.

• Build a student-facing dashboard to allow self-monitoring of risk scores.

Long-Term Strategy

• Use clustering persona profiles to design targeted learning programs.

• Track intervention outcomes to measure the system's effectiveness.

• Expand dataset with library usage, hostel data, and attendance trends over time.

This report was generated automatically by the Smart Campus Analytics System. All ML models were trained and evaluated on 500 CS student records. For queries, contact the Analytics Team.